

SmartShelf : A Smart Library

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1 Project Scope

The role of libraries starts with serving as an integral support system for knowledge sharing and community development, offering access to books, information, and knowledge resources to all age groups of society. In a community context, there is an increasing complexity in effectively managing the library due to multiple users, an ever-increasing collection of resources, and regular activities of borrowing and returning. Manually handling, or using a semi-manual system, in library management is often time-consuming, error-prone, and inefficient, and hence results in problems such as inaccuracies, lag in updates, loss of resources, and inappropriateness in generation of reports.

To overcome these difficulties, **SmartShelf** has been developed as an integral **Database Management System (DBMS)**, automating the main activities in managing a community library. It plans to offer an effective, efficient, and convenient interface among library users, librarians, and administrators, while promoting accuracy, consistency, and simplicity in accessing the database.

- **Students/Members** can register and log into the system using secure credentials. They can browse the library catalog, search for books by title, author, or category, check availability status, and issue or reserve books online. Users can also view their borrowing history, due dates, and any applicable fines, ensuring transparency and convenience.
- **Librarians** manage the day-to-day operations of the library. They are responsible for adding new books, updating book details, issuing and returning books, and managing reservations. Librarians can also monitor overdue/damaged books and ensure that library rules are enforced efficiently.
- **Administrators** oversee the entire system, ensuring smooth functioning and data integrity. They manage user roles, monitor system usage, generate reports, add/remove book entries and maintain overall control of the library database.

2 Detailed Overview of the Functionality

2.1 User / Customer Management

- **Signup:** It allows the user to sign up into the app by creating an account using valid credentials, including unique email ID, phone number, and secure password.
- **Login:** The system allows access only to registered members who can log in using the provided credentials.
- **Book Catalog:** An organized and easily searchable catalog of all available books, arranged by genre, authors, and topics.
- **Search Functionality:** This feature enables users to search for books easily based on key words, for example, title, author, or ISBN.
- **Book Issue and Reservation:** A user can issue an available book or make a reservation request for a book that is currently unavailable.
- **Borrowing History:** Users can view their previously issued books along with issue and return dates.
- **Membership:** Users will be able to buy membership, which will enable them to borrow more than one book at a time and will also provide extended dates.
- **Due Date and Fine Tracking:** This feature shows the due dates of the issued books, with fines, if any, for delayed returns.
- **Logout:** Enables users to logout of the system safely.

2.2 Librarian Management

- **Login:** Librarians login by using the authorized credentials.
- **Manage Books:** Manage adding, editing, or deleting books from the library database.
- **Issue and Return Management:** Book issue and return processes, with its real-time availability status, are being handled.
- **Reservation Handling:** Approve or deny reservations according to the availability of books.
- **Overdue Monitoring:** Monitor books that have become overdue and notify users accordingly.
- **Missing/Damaged books:** Requests for the fine for missing or damaged books.

2.3 Admin Management

- **Login:** Secure login facility for administrators.
- **User Management:** Handling user/member and librarian profiles.
- **System Monitoring:** The monitoring of the usage of the libraries, books issued, and the performance of the system.
- **Supplying Books:** Provides new entries of books into the library, which are collected from the publishing houses. Also can removes entries of books.
- **Reports and Statistics:** The system will enable report creation on book circulation, most issued book records, overdue records, and user activity.
- **Data Maintenance:** Ensure data consistency, backup, and system integrity.

3 Technical Requirements

The following technologies will be used for the development and implementation of the Library Management System:-

- **Database:** MySQL
- **Programming Languages:** Python, HTML, CSS, JavaScript
- **Frontend and Backend:** ReactJS, Flask

4 Contribution

We have collectively decided upon this topic for the DBMS Project and all have contributed equally for the same.